

**DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 330

Homework Set 1

Problem 1 (20%) In a heat exchanger experiment, the following data have been collected on a hot water stream: mass flow rate = 4.0 ± 0.1 kg/min, inlet temperature = $56.0 \pm 0.05^\circ\text{C}$, outlet temperature = $42.5 \pm 0.05^\circ\text{C}$. Given that the specific heat of water is $1.0 \text{ kcal}/(\text{kg}^\circ\text{C})$, determine **(a, 7%)** the heat transfer rate from the hot water stream, and **(b, 13%)** the uncertainty in the heat transfer rate due to measurement uncertainties.

Problem 2 (20%) In a study of free convective heat transfer from a heated metal cylinder with a diameter (D) of 0.010 m and a heat transfer area (A_s) of $2.20 \times 10^{-3} \text{ m}^2$, the following data have been collected:

- Voltage applied across the metal cylinder, $V = 16.0 \pm 0.4 \text{ volt}$
- Electric current flowing through the metal cylinder, $I = 3.6 \pm 0.2 \text{ A}$
- Surface temperature of metal cylinder, $T_s = 760 \pm 2 \text{ K}$
- Ambient air temperature, $T_a = 296 \pm 1 \text{ K}$

Given that power input to the metal cylinder is given by

$$Q_{in} = V \cdot I \quad (1)$$

and the free convective heat transfer rate from the metal cylinder is given by

$$Q_{nc} = H_{nc} A_s (T_s - T_a) \quad (2)$$

where the heat transfer coefficient for natural convection (H_{nc}) can be found by:

$$H_{nc} (\text{W}/\text{m}^2 \cdot \text{K}) = 1.32 [(T_s - T_a)/D]^{0.25} \quad (3)$$

determine **(a, 2%)** Q_{in} , **(b, 4%)** Q_{nc} , **(c, 5%)** uncertainty in Q_{in} , and **(d, 9%)** uncertainty in Q_{nc} .

Problem 3 (20%) In a study of the head loss coefficient of an expansion valve, the following set of data has been collected:

- Inlet valve diameter = 0.0225 m , Outlet valve diameter = 0.0296 m
- Volumetric flow rate of water through the valve = $(4.80 \pm 0.10) \times 10^{-4} \text{ m}^3/\text{s}$
- Manometer readings: at inlet valve = $236 \pm 2 \text{ mm H}_2\text{O}$, at outlet valve = $249 \pm 2 \text{ mm H}_2\text{O}$

Given that the mechanical energy balance across the expansion valve is given by

$$\frac{P_2 - P_1}{\rho} + \frac{(v_2^2 - v_1^2)}{2} = -K_d \frac{v_1^2}{2} \quad (4)$$

where P is the fluid pressure, v is the fluid velocity, ρ is the fluid density ($= 1.0 \text{ g}/\text{cm}^3$), K_d is the head loss coefficient, and subscripts 1 and 2 denote inlet and outlet conditions, respectively, determine **(a, 7%)** the head loss coefficient of the expansion valve and **(b, 13%)** its uncertainty.

Problem 4 (20%) In a hot- and cold-water mixing experiment, the following data have been collected on the hot and cold water streams: hot water flow rate ($\dot{V}_h$) = $1.00 \pm 0.05 \text{ L/s}$; hot water temperature (T_h) = $58.0 \pm 0.05^\circ\text{C}$; cold water temperature (T_c) = $15.0 \pm 0.05^\circ\text{C}$. By rearranging

the steady state energy balance, the theoretical cold water flowrate ($\dot{V}_c$) can be calculated by:

$$\dot{V}_c = \frac{T_h - T_{mix}}{T_{mix} - T_c} \dot{V}_h \quad (5)$$

If it is desired that hot and cold water streams are mixed to give a set-point temperature of 40.0 ± 0.05 °C, determine **(a, 7%)** the theoretical cold flow rate, and **(b, 13%)** the uncertainty in the theoretical cold flow rate due to measurement uncertainties.

Problem 5 (20%) The heat (or enthalpy) of a reversible or equilibrium reaction is given by

$$\Delta H_r = \frac{-R \ln \left(\frac{K_{T_2}}{K_{T_1}} \right)}{\left(\frac{1}{T_2} - \frac{1}{T_1} \right)} = \frac{-RT_1 T_2}{(T_1 - T_2)} (\ln K_{T_2} - \ln K_{T_1}) \quad (6)$$

Given the following set of data, determine **(a, 3%)** the heat of the reaction and **(b, 17%)** its uncertainty:

$T_1 = (298.15 \pm 0.2)$ K, $K_{T_1} = 0.0322 \pm 0.0008$, $T_2 = (353.15 \pm 0.4)$ K, $K_{T_2} = 0.473 \pm 0.008$.

Hint: For $y = a \ln(bx)$, where a and b are constants, $\delta y = a \frac{\delta x}{x}$, where δ denotes uncertainty.